



Abhinav Shukla

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Bachelor of Technology(Hons.)

in Computer Science & Engineering (AI)

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GitHub

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SUMMARY

Innovative and results-driven Computer Science (AI) student with a strong foundation in programming, machine learning, and leadership. Committed to advancing research and innovation in AI and machine learning, with a focus on solving complex, real-world problems. Passionate about exploring emerging technologies, contributing to academic communities, and fostering knowledge exchange through technical leadership and scholarly initiatives.

EDUCATION

Degree	Institute	Board / University	CGPA/Percentage	Year
B.Tech(H) CSE (AI)	University Teaching Department, Chhattisgarh Swami Vivekanand Technical University	CSV TU Bhilai	6.71(3rd sem)	2023-2027
Senior Secondary	Lions' DAV Public School	CBSE	74.8%	2022
Matriculation	Lions' DAV Public School	CBSE	81.2%	2020

PROJECTS

Explainable GeoAI (XGBoost + SHAP + GeoShapley + MGWR)

May 2025

Built an end-to-end spatial ML pipeline to predict U.S. county vote-share ($R^2 0.90$) using FLAML-tuned XGBoost and GeoPandas, with integrated Tree SHAP attributions and 500-sample bootstrap uncertainty. Extended interpretability via GeoShapley's decomposition (base, location, feature, interaction effects) and benchmarked against MGWR local coefficients. Conducted spatial fairness audits across demographic quartiles and delivered an interactive Streamlit dashboard (Folium + Plotly) for on-the-fly exploration.

[Github](#)

– Skills: XGBoost, FLAML, SHAP, GeoShapley, MGWR, GeoPandas, Folium:

Full World Model (Vision + Memory + Controller)

Apr 2025

Developed a Full World Model architecture using Convolutional Networks, LSTM-based Memory, and a Controller to autonomously drive in the CarRacing-v3 environment. Achieved successful rollout with latent state prediction, reinforcement learning rewards, and automated performance evaluation with video recording.

[Github](#)

– Skills: Deep Learning, Model-Based RL, CNN, LSTM, PyTorch, Environment Modeling, Gymnasium:

Fine-Tuned LLaMA 3.2 (3B Model)

Apr 2025

Fine-tuned LLaMA 3.2 (3B parameters) using Parameter-Efficient Fine-Tuning (PEFT) and LoRA techniques with Unsloth, optimizing training pipelines via Hugging Face Datasets. Achieved a 22% reduction in computational overhead while improving model response quality by 15%, using SFTTrainer for structured instruction tuning.

[Github](#)

– Skills: NLP, LLM Fine-Tuning, Hugging Face, LoRA, PEFT, Unsloth, SFTTrainer:

Cold Email Generator using GenAI

Nov 2024

Developed an AI-powered tool that automates personalized cold email generation for software and AI service companies. Integrated LLaMA 3.1 with ChromaDB and LangChain to match resumes with job descriptions and generate emails with 90% relevance accuracy. Built a Streamlit app for a user-friendly, zero-code interface, improving client outreach speed by 40%.

– Skills: Generative AI, LangChain, ChromaDB, LLaMA 3.1, Streamlit:

EcoLens — AI for E-Waste Management

Jan 2024

Designed EcoLens, an AI-driven application combating global e-waste through intelligent computer vision. Using CNN models, it identifies, classifies, and suggests eco-friendly disposal methods for electronic waste with 85% classification accuracy. The tool educates users on sustainable practices, supporting UN SDG 12 (Responsible Consumption and Production).

[Github](#)

– Skills: Computer Vision, Deep Learning, Sustainability Tech, Image Classification:

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Python, Rust, Julia, Bash, R
- **Machine Learning & AI Frameworks:** TensorFlow, PyTorch, Scikit-Learn, Keras, OpenCV, Gymnasium
- **Deep Learning & Reinforcement Learning:** CNN, LSTM, DDPG, Evolution Strategies, World Models, Policy Gradient Methods
- **Data Science & Analytics:** NumPy, Pandas, Seaborn, Matplotlib
- **AI & LLM Development:** LangChain, ChromaDB, Hugging Face Transformers
- **Databases:** PostgreSQL, SQLite, ChromaDB, Firebase
- **Cloud Computing & DevOps:** Git, GitHub Actions, CI/CD Pipelines, Prometheus, Grafana, ELK Stack, Cron Jobs
- **Monitoring & Automation:** Structured Logging, Instrumentation, JSON Reporting, Push/Pull Metrics, Distroless Containers, Workflow Orchestration
- **Development Tools:** Jupyter, VS Code, Sublime, PowerShell, CLI Utilities in Rust
- **Operating Systems:** Windows, Linux, macOS, TailsOS, Android

PAPERS PUBLISHED/PRESENTED

- **EcoLens : A Visionary Solution For E-Waste Management Through Intelligent Imaging,**
- ,Presented at NITRR Alumini Association Seminar

POSITIONS OF RESPONSIBILITIES

- **Head of Technical Council**,Programmers’ Paradise, Dept. of CSE, CSVTU Bhilai Mar 2025 – Present
 - Leading technical initiatives across all chapters; strategizing collaborations, events, and skill development programs.
 - Mentoring council members to sustain leadership, innovation, and coding culture within the university.
- **President**,Programmers’ Paradise Mar 2024 – Mar 2025
 - Organized 20+ workshops and hackathons; made the club Chhattisgarh’s most active coding community.
 - Secured MoUs with Swapso.io and Altair to launch Blockchain, Web3, and AI initiatives at CSVTU.
 - Established MATLAB and Blockchain Web3 Chapters; expanded technical learning pathways.
 - Managed IEEE ICECCT 2024 technical operations and collaborated with TPO for industry outreach.
- **TPO Representative**,Training and Placement Office, CSVTU Apr 2024 – Jul 2024
 - Revamped the TPO student body structure, boosting efficiency and student-industry engagement.
 - Coordinated recruitment drives and promoted coding events to enhance placement readiness.
- **Technical Lead**,IEEE ICECCT 2024 Jun 2024
 - Directed technical execution for the international conference; hosted keynote sessions and panels.
- **House Captain**,DAV Public School May 2021 – Apr 2022

CERTIFICATIONS

- **Fundamentals of Reinforcement Learning** University of Alberta
- **Introduction to Deep Learning and Neural Networks with Keras** IBM
- **Machine Learning With Python** IBM
- **Rust for DevOps** Duke University
- **Linear Algebra for Machine Learning and Data Science** DeepLearning.ai
- **Introduction to Neural Networks and Pytorch** IBM
- **Data Engineering with Rust** Duke University
- **Developing Explainable AI (XAI)** Duke University
- **Deep Learning with Keras and Tensorflow** IBM
- **Deep Learning with PyTorch** IBM

KEY COURSES TAKEN

- Programming Software Development:**Data Structures with C, Object-Oriented Programming (C++), Operating Systems, Database Management Systems
- Artificial Intelligence Data Science:** Artificial Intelligence: Principles Applications, Python for Data Science, R for Data Science, Data Visualization
- Mathematics Algorithms:**Discrete Structures, Probability Statistics, Engineering Mathematics I II, Analysis and Design of Algorithms, Theory of Computation
- Computer Systems Networks:**Computer Organization and Architecture, Computer Networks
- Electronics Logic Design:**Foundations of Electronics Engineering, Digital Design and Logic

EXPERIENCE

- **Dr. SPM International Institute of Information Technology, Naya Raipur**
AI/ML Research Intern

Apr 2025 - Present
Hybrid

* Designed and implemented a robust real-time people tracking system that dynamically adapts to changing scenes, delivering high accuracy and performance. Developed an automated tuning and evaluation framework to continuously optimize and validate tracking results.
- **GirlScript Summer of Code**
Open Source Contributor

Summers 2024
Online

* Worked on many open source projects
- **Social Summer of Code**
Open Source Contributor

Summers 2024
Online

* Worked on many open source projects

ACHIEVEMENTS

- **Hackerrank 5-Star Achiever**,Earned a 5-star rating in competitive programming on Hackerrank.

2023
- **Best Paper** ,Received prize on the basis of our paper presentation and project by NITRR AA seminar

2024
- **Top Ranker**,Was rank 19 in International Informatics Olympiad 2019

2019